JOSEANE PEREIRA

Computer and Telematics Engineer, MSc

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PROFILE

MSc graduate in Computer and Telematics Engineering (University of Aveiro) with hands-on experience across network systems, AI/LLM applications, and full-stack software development. Thesis work centered on integrating a commercial network-management platform with an open-source 5G core, spanning protocol-level programming, systems integration, and real-time control. Additional experience includes building a RAG-based chatbot on the OpenAI API and a cloud-deployed REST API on AWS. Looking to bring this cross-disciplinary foundation to an entry-level engineering role.

EDUCATION

MSc in Computer and Telematics Engineering, University of Aveiro (2024–2026, thesis grade 16/20)
Thesis: *"Mobile Network Systems Orchestration"* (advisors: Prof. Daniel Corujo, Prof. Flávio Meneses)

BSc in Computer and Informatics Engineering, University of Aveiro (2021–2024)

EXPERIENCE

Mobile Network Systems Orchestration, MSc Thesis

- Built an external orchestration layer linking a commercial network-management platform (DataMiner) to an open-source 5G core (Open5GS) through its native metrics, database, and control interfaces: hands-on systems integration across heterogeneous platforms.
- Extended core-network internals: implemented a Python sidecar and modified the Open5GS AMF in C to expose a REST endpoint for signalling behaviour the core didn't natively support, requiring protocol-level programming and comfort working inside a large existing C codebase.
- Built a real-time, telemetry-driven traffic-shaping service (Prometheus metrics, Linux tc/HTB) on the network's user-plane, and delivered the whole system as a reproducible, documented artifact published on GitHub.

Team Lead, Deloitte Tech Experience · Lisbon · 2024
Led a 5-person team building a document-grounded chatbot on the OpenAI API, implementing retrieval-augmented generation (embeddings + vector search) to constrain every answer to a limited internal document set.

AWS Product API, Sky Careers Technical Assessment · 2024
Designed and built a product-management REST API in C# on AWS as a 1-week technical interview challenge: data modeling, endpoint design, and cloud deployment.

Tech across these: Python, C, C#, OpenAI API, embeddings/vector search, Open5GS, DataMiner, MongoDB, Prometheus, AWS, Linux (tc/HTB), REST APIs

SKILLS

AI / LLM: OpenAI API integration, prompt engineering, retrieval-augmented generation (embeddings, vector search).

Networks & Systems: TCP/IP, UDP, routing, Wireshark; firewalls & VPNs; network/service virtualization; heterogeneous & mobile network architectures.

Programming & Development: Python, C/C++, C#, Java, JavaScript, React; Git; SQL; REST API design; AWS (hands-on); agile methodologies; Linux (Arch/CachyOS daily driver, Wayland/Niri compositor config).

Parallel & High-Performance Computing: pthreads, MPI, CUDA; architecture simulation and optimization.

Embedded Systems & Signal Processing: Sensors/actuators; C/C++ for hardware; digital filters, DSP; MATLAB & Simulink; analog/digital electronics; RF/wireless fundamentals.

PROJECTS

Also on GitHub (github.com/Joseane772): compiler construction (Java), interactive web/game projects (JavaScript, HTML).

LANGUAGES

Portuguese: Native (C2) | English: Advanced (C1)